Poula Kerolos Labeb

Cairo, Egypt | +20-12-77925517 | poulakerolos7@gmail.com | github.com/Poulakerolos |
linkedin.com/in/poula-kerolos-63614836a

PROFESSIONAL SUMMARY

Electronics and Electrical Communications Engineering student at Cairo University (GPA: 3.82), specializing in analog and mixed-signal electronics with hands-on depth in digital design, embedded Linux, and systems programming. Built and published a Verilog RV32I processor, a userspace filesystem in C, and custom Unix shells.

EDUCATION

Cairo University - Faculty of Engineering *2022 - 2026 (Expected)*

B.Sc. Electronics and Electrical Communications Engineering | Cumulative GPA: 3.82 | Last Semester GPA: 3.67

Relevant coursework: Circuits I & II, Electronics, Logic Design, Data Structures, ODE, Probability & Statistics, Optics

SELECTED PROJECTS - PUBLISHED ON GITHUB

Filesystem From Scratch - C *GitHub: filesystem-from-scratch*

- Implemented a userspace filesystem with superblock, inode, bitmap free-space tracking, nested directories, file I/O, and host-file import/export.
- Added recoverable deletion, inline small-file storage, and separate-process persistence and recovery tests.

Single-Cycle RV32I Processor - Verilog *GitHub: verilog-projects*

- Integrated program counter, instruction/data memories, register file, decoders, ALU, branch logic, program image, and testbench into a single-cycle processor.
- Published supporting RTL modules including a 4-bit ALU, dual-port RAM, clock divider, LIFO stack, FSM sequence detectors, and arithmetic shift register.

Custom Unix Shells - C / POSIX *GitHub: custom-shells*

- Built a three-tier progression from static built-ins to dynamic command parsing with fork/execvpe and shell-managed local/global variables.
- Published companion Unix-style utilities using low-level file APIs and a clearly labeled team assembly string-library artifact.

WORKSHOPS & TRAINING

Digital Design Workshop - IEEE Cairo University Student Branch *2025*

Verilog HDL | ModelSim/Questa simulation and testbench verification

Analog Design Workshop - IEEE Helwan University Student Branch *2025*

Cadence Virtuoso | Certificate awarded; MOSFET fundamentals, amplifier topologies, biasing, schematic entry, and simulation.

RESEARCH & ACADEMIC ACTIVITIES

Autonomous Vehicle Collision Avoidance - Research Day Presenter *2025*

Limit Breakers Team | Supervised by Dr. Samah El-Tantawy, Cairo University

- Co-developed a LiDAR-based collision-avoidance demo car using Kalman filtering, LQR optimal-control braking, and dynamic time-to-collision thresholds.
- Presented the system to judging panels and in a Research Day poster session.

TECHNICAL SKILLS

Programming & HDL: C, Verilog HDL, C++, Python | **Tools:** ModelSim/Questa, Cadence Virtuoso (familiar), Arduino IDE, Multisim, KiCad, LTspice

Domains: Analog and Mixed-Signal Electronics, Embedded Linux, Systems Programming, Digital Design, Embedded Systems, Autonomous Systems

LANGUAGES

Arabic (Native) | English B2 - Ministry of Defense Language Institute | German A1 - Ministry of Defense Language Institute